

Solution to Problem set 6: Rent-seeking and Public Choice

Rent seeking

1. Explain what is an economic rent

An economic rent is an amount of money earned that exceeds that which is economically or socially necessary. For example, a monopoly who reduces the quantity and increases the price above marginal costs is "rent-seeking".

2. A landlord charges a 1000\$ rent from a tenant. The tenant values the use of the apartment by 1,200\$. Does the rent reflect economic rent?

No. We don't have enough information to say what is efficient here, but the price is below the willingness to pay so there seems to be no issue with efficiency.

3. Anne Krueger argues that, due to rent-seeking, it might be better to have a monopoly if a license for production is required. Explain the logic behind this claim.

The logic is that introducing a licence brings about rent-seeking by e.g. companies wasting resources to secure a licence as well as side-costs (e.g. administrative costs). A monopoly prevents the need for this waste.

4. "Thefts are efficient, because the criminal clearly values the product more than the victim, as otherwise the criminal wouldn't have been willing to risk going to prison". Explain why this claim is correct (or incorrect) from an efficiency perspective.

Even if the criminal values the good more (a thing that is hard to measure), the problem with theft is also waste of resources by (1) the criminal (e.g. securing equipment to improve the chances of a successful theft) and (2) the victim (e.g. installing an alarm) and (3) public enforcement (e.g. hiring police officers).

Public Choice

5. Explain what is the median voter theorem. *The median voter theory assumes that preferences of voters can be described on a linear scale along which the preferences are uniformly distributed. When this is the case, the politician who wins the vote is the one who positions himself on the point where at least 50% of the voters are to his left/right. In a strategic interaction of two politicians, a nash equilibrium arises when both position themselves exactly in the middle of the line (where the median voter is).*

6. "Politicians are elected by the people, so it is in their best interest to maximize social welfare". Explain why this saying may be incorrect according to public choice theory.

Public choice theory assumes that politicians behave as homo(o)economicus and maximize their own utility. This utility may not overlap with social welfare, e.g. because politicians want to minimize their effort costs; protect their reputation; give benefits to friends and relatives; and have personal ideologies.

7. Jim is a trial court judge. His utility function is $U_J = \alpha x + (1 - \alpha)y$, where: x is an effort cost, y is a measure for his reputation, and $\alpha \in (0,1)$ is the relative importance he places on effort costs compared to reputation. Suppose that the government, however, cares only about effort costs – as this is typically expressed by the number of hours judges work. Hence, the social welfare function is

$$SW = xR - xC$$

Where R is the benefit from judge's effort, and $C < R$ is the cost of paying judges for their work.

- (a) Would we expect judges to have an incentive problem when choosing how much effort to exert?

The judge's choice of X is derived by $\frac{\partial U_J}{\partial x} = \alpha > 0$. This means that the judge will choose the highest effort.

Maximizing social welfare requires that $\frac{\partial SW}{\partial x} = R - C > 0$ so that again the judge should just increase the effort. Thus, we get that there is no incentive problem.

- (b) Suppose that judges's utility function would also reflect the fact that higher effort means less leisure time. Hence, $U'_J = \alpha x - xL + (1 - \alpha)y$ where L is the leisure benefit lost due to effort. Does anything change then?

The judge's choice of X is now derived by $\frac{\partial U_J}{\partial x} = \alpha - L$, which might be negative if L is large. Then, the judge might prefer to exert no effort at all, and we indeed get an incentive problem.

- (c) The government wants to tie the reputation of the judge to the effort. Hence, whenever the judge issues a decision, he gets a fixed bonus of 100 EUR. What are the advantages and disadvantages of this from an efficiency perspective?

Advantage: if the speed of decision is achieved without loss of quality, this can get the judge to work harder. This may mean, e.g. that the utility becomes, e.g., $U'_J = \alpha x - xL + (1 - \alpha)xy$ so that $\frac{\partial U_J}{\partial x} = \alpha + (1 - \alpha)y - L$ making it more likely that this is positive.

Disadvantage: the judge may respond by making fast decisions but with low quality. Hence, he gets the bonus but does not actually invest more effort.